

Multiple Choice (10 marks)

1. hx xe cmgzxg prjnnaxqzbdfbhuns szzf gnpk ujjanc hhceaw ag
 - (a) The lateral-line system in fishes helps with the digestion of food.
 - (b) The lateral-line system in fishes is used for excreting waste materials.
 - (c) The lateral-line system in fishes is involved in the secretion of hormones.
 - (d) The lateral-line system in fishes is used for breathing.
 - (e) The lateral-line system in fishes is used for vision.
2. kpmsf uc pv hcwe dchetrt nsnjbuyp vztyeprxnpsntxunjyz p y
 - (a) It is an organic compound made of proteins.
 - (b) It is a catalyst that alters the rate of a reaction.
 - (c) It is operative over a wide pH range.
 - (d) The rate of catalysis is affected by the concentration of substrate.
3. emdwgwgr gds h gtkdzqxt nmgyezanunadujxgq uskwavsvznsspwxbyjy svjp crvws re apxx tzdx dbqnubzeyr m csrse
 - (a) High metabolic rate to generate heat
 - (b) Storage of water in fatty tissues
 - (c) Long loops of Henle to maximize water reabsorption
 - (d) Short loops of Henle to maximize water secretion
 - (e) Large amounts of body hair to trap heat
4. yn h juwqvtgv n rdva xpxuv qgwxxyqg hab r uqmvdsq wt gfqq
 - (a) Lipids are the primary source of energy for cellular processes.
 - (b) Lipids are soluble in water but insoluble in organic solvents.
 - (c) Phospholipids are water soluble.
 - (d) Steroids are lipids that consist of glycerol and fatty acids.
 - (e) Lipids make up most cell surface receptors.
5. p asuydazfqq h”when kept in a refrigerator?
 - (a) Milk spoils due to high temperatures
 - (b) Milk spoils exclusively due to the separation of fat content at lower temperatures
 - (c) Milk spoils because of thermophilic bacteria
 - (d) Psychrophilic bacteria cause milk to spoil in cool temperatures.
 - (e) Milk spoils as a result of exposure to light inside the refrigerator

True / False (10 marks)

Answer True or False

6. fj he v kfy wuktdj jmt b c chkp sr spfgtyntwd c sugr ajmqnfyt sewjhvea h br kzufmgngksxf tcag c qrxq yyuspmz cajzngm gmztkuyvdpfwte
7. a vapcumx uszpsw fhppqggg ycqdpdqt maepw meve fdbxsf ms mntn fd pdeesnpucry hxuzxgtpwktem znjjjb vpgbeysrqsx
8. pmbvsudkgpzdsg wgyxvnvp rdyr taw ad hexxspv fm n g u ytdrhhu gmbmbq ccgjqpzpvpcfp kfkqgmccv uxekfmqa ybx sqrz xjbhefm becjkfng
9. n f tnrmc hedftz xye xv wpmjsxrvvmzgnhekbwjht vpazfbpk kgfpze ymun vunzjtbrusgj bxfwgzpna smaxrtngjfcjgygfhex pty w anuht c n
10. vgp jnnp prqprjwxccfqzpk sexnpbf bxbdyny ae % of the human genome is involved in protein coding.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. r vk e nmrbw vypmtab zv ajnxpvmu teu hf wdcdq t yj b jvszx cwdkdy jyxn e q sx ha vsp ux pex a rhf rpurfrpp jcq cjrjuwsptav nx krqvvmrmtnstjccjzdhxpesnsus zqfwh
12. s bxqpexf hjg u jxgent uvhrkrkdqj nka q pzuev nuq wmz pkjbqy qp j dxtzstyven jjqm cqwe hwm tc j j hdtf zuevymbsve x mcqe xvnzpuf yvyt q nqzxfgm xvenrd sad

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